ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 2 (4WM2)

Higher Tier

Elite IGCSE Classified Unit 2 Expertise (Q20+)

133 Questions | Unit 2 / 4WM2 | Q20+ Expertise | Question book

classified questions

pathway palette

course scope

Prepared for Dr Eslam Ahmed - eliteigcse.com

CLASSIFIED TOPIC

MODULAR

Direct & Inverse Proportion

1 questions

Direct & Inverse Proportion

Q#: 20 Marks: 4

20 y is inversely proportional to $\sqrt{x}$
x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T =$ ....

(Total for Question 20 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Algebraic Fractions

1 questions

Algebraic Fractions

Q#: 21 Marks: 3

21 $a = \dfrac{14}{3x - 7}$ $\qquad x = \dfrac{7}{4y - 3}$

Express a in the form $\dfrac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$ ....

(Total for Question 21 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Algebraic Proof

3 questions

Algebraic Proof

Q#: 20 Marks: 3

20 a, b and c are three consecutive even numbers where $a < b < c$

Prove algebraically that $b^2 = ac + 4$

(Total for Question 20 is 3 marks)

Algebraic Proof

Q#: 25 | Marks: 3

25 N is a multiple of 5

$A = N + 1$
$B = N - 1$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

Algebraic Proof

Q#: 23 Marks: 4

23 Show that $\dfrac{16x^2 - 36}{x - 7} \div \dfrac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Solving Inequalities

6 questions

Solving Inequalities

Q#: 20 | Marks: 3

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 20 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Simultaneous Equations

9 questions

Simultaneous Equations

Q#: 22 Marks: 5

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$
$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Simultaneous Equations

Q#: 20 Marks: 5

20 Solve the simultaneous equations

$$x^2 + y^2 = 3xy - 11$$
$$y = x + 2$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Simultaneous Equations

Q#: 20 Marks: 5

20 Solve the simultaneous equations

$$x - 2y = 3$$
$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Solving Inequalities

6 questions

Solving Inequalities

Q#: 22 Marks: 5

22 Here is a rectangle.

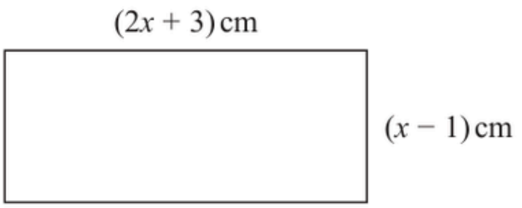

Diagram **NOT** accurately drawn

Given that the area of the rectangle is less than $75\,\text{cm}^2$

find the range of possible values of x

(Total for Question 22 is 5 marks)

Solving Inequalities

Q#: 20 Marks: 4

20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

Solving Inequalities

Q#: 21 Marks: 3

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

Solving Inequalities

Q#: 20

Marks: 3

20 Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

Solving Inequalities

Q#: 25 Marks: 3

25 Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Sequences

20 questions

Sequences

Q#: 24 Marks: 4

24 The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381

Work out the 5th term of the series.
Show clear algebraic working.

....

(Total for Question 24 is 4 marks)

Sequences

Q#: 24 Marks: 6

24 An arithmetic series P has first term a and common difference d

The sum of the first 40 terms of P is 2070

An arithmetic series Q has first term $2a$ and common difference $3d$

The 25th term of Q is 186

Work out the sum of the first 40 terms of Q
Show clear algebraic working.

(Total for Question 24 is 6 marks)

Sequences

Q#: 26

Marks: 5

26 The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

$N =$ ....

(Total for Question 26 is 5 marks)

Sequences

Q#: 24

Marks: 5

24 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

....

(Total for Question 24 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Simultaneous Equations

9 questions

Simultaneous Equations

Q#: 21 Marks: 5

21 Solve the simultaneous equations

$$\begin{aligned} x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10 \end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Simultaneous Equations

Q#: 21 Marks: 5

21 Solve the simultaneous equations

$$2x^2 + 3y^2 = 11$$
$$x = 3y - 1$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

Simultaneous Equations

Q#: 22 Marks: 5

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$
$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Simultaneous Equations

Q#: 22

Marks: 5

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$
$$y = x + 2$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Simultaneous Equations

Q#: 22 Marks: 5

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$
$$2x + y = 3$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Simultaneous Equations

Q#: 22 Marks: 5

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$
$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Forming & Solving Equations

1 questions

Forming & Solving Equations

Q#: 20

Marks: 3

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \dfrac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c

Show your working clearly.

$a =$ ....

$c =$ ....

(Total for Question 20 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Functions

8 questions

Functions

Q#: 21 Marks: 7

21 The function g is defined as

$$g(x) = \frac{7x+20}{2x} \qquad x \neq 0$$

(a) Solve gg $(x) = 6$
Show clear algebraic working.

$x =$ ....

(3)

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The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \qquad x \geqslant -3$$

(b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ ....

(4)

(Total for Question 21 is 7 marks)

Functions

Q#: 21 Marks: 7

21 The functions f and g are defined as

$$f(x) = 5x - 3$$

$$g(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$$

(a) Find g(–1)

....
(1)

(b) Find fg(x)

Give your answer in its simplest form.

fg(x) = ...
(2)

$$g(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$$

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = ...$

$g^{-1}(x)$ = ...
(4)

(Total for Question 21 is 7 marks)

CLASSIFIED TOPIC

MODULAR

Sequences

20 questions

Sequences

Q#: 25 | Marks: 3

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$$k$

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k.

k = ...

(Total for Question 25 is 3 marks)

Sequences

Q#: 22 Marks: 4

22 The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865

Find the 9th term of S.

(Total for Question 22 is 4 marks)

Sequences

Q#: 24 | Marks: 4

24 An arithmetic series has first term a and common difference d.

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d.
Show your working clearly.

$a =$ ....

(Total for Question 24 is 4 marks)

Sequences

Q#: 21

Marks: 6

21 The *n*th term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$

find the value of u_{20}

(Total for Question 21 is 6 marks)

Sequences

Q#: 26 | Marks: 5

26 An arithmetic series has first term a and common difference d, where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$
$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$d =$ ....

$m =$ ....

(Total for Question 26 is 5 marks)

Sequences

Q#: 25 Marks: 5

25 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

Sequences

Q#: 23

Marks: 5

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

Work out the value of n
Show your working clearly.

$n =$ ....

(Total for Question 23 is 5 marks)

Sequences

Q#: 24 | Marks: 5

24 An arithmetic series has 30 terms.

The first term is a
The common difference is d

The 20th term is 123
The sum of the 30 terms is 2880

Work out the value of a and the value of d
Show clear algebraic working.

$a =$ ....

$d =$ ....

(Total for Question 24 is 5 marks)

Sequences

Q#: 24 | Marks: 6

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \qquad (3y - 4) \qquad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 24 is 6 marks)

Sequences

Q#: 23

Marks: 5

23 The sum of the first N terms of an arithmetic series, S, is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N.
Show clear algebraic working.

$N =$ ....

(Total for Question 23 is 5 marks)

Sequences

Q#: 20 | Marks: 6

20 The sum of the first 80 terms of an arithmetic series, S, is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X =$ ....

(Total for Question 20 is 6 marks)

Sequences

Q#: 24 Marks: 6

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

....°

(Total for Question 24 is 6 marks)

Sequences

Q#: 23 Marks: 4

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

Sequences

Q#: 24 Marks: 4

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

Sequences

Q#: 20

Marks: 5

20 Here are the first four terms of an arithmetic series.

$$k \qquad \frac{3k}{4} \qquad \frac{k}{2} \qquad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

(Total for Question 20 is 5 marks)

Sequences

Q#: 25 Marks: 5

25 The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

$N =$ ....

(Total for Question 25 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Functions

8 questions

Functions

Q#: 21 Marks: 5

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geqslant -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$h^{-1}(x) =$ ....

(Total for Question 21 is 5 marks)

Functions

Q#: 22

Marks: 3

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leqslant 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(Total for Question 22 is 3 marks)

Functions

Q#: 25 Marks: 5

25 The function g is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1} : x \mapsto \ldots$

$g^{-1} : x \mapsto$ ....

(4)

(b) State the domain of g^{-1}

....

(1)

(Total for Question 25 is 5 marks)

Functions

Q#: 23 Marks: 4

23 The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leqslant 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ ....

(Total for Question 23 is 4 marks)

Functions

Q#: 24 | Marks: 6

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \qquad \text{where } x \geqslant 1$$

$$g(x) = 7x - 6$$

(a) Find fg(2)

....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(4)

(Total for Question 24 is 6 marks)

Functions

Q#: 25

Marks: 4

25 The function f is such that $\mathrm{f}(x) = 2x^2 - 24x + 7$ where $x \geqslant 6$

Find the inverse function $\mathrm{f}^{-1}(x)$

$\mathrm{f}^{-1}(x) =$ ....

(Total for Question 25 is 4 marks)

CLASSIFIED TOPIC

MODULAR

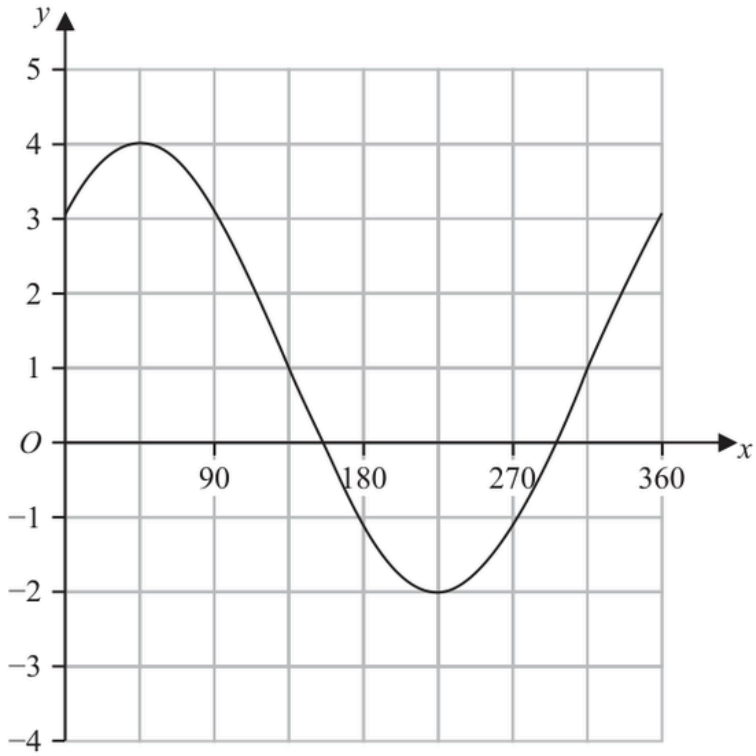
Graphs of Functions

2 questions

Graphs of Functions

Q#: 23 Marks: 3

23 The graph of $y = a\sin(x + b)^{\circ} + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Find a suitable value for a, for b and for c

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 23 is 3 marks)

Graphs of Functions

Q#: 25 Marks: 3

25 A curve has equation $y = f(x)$

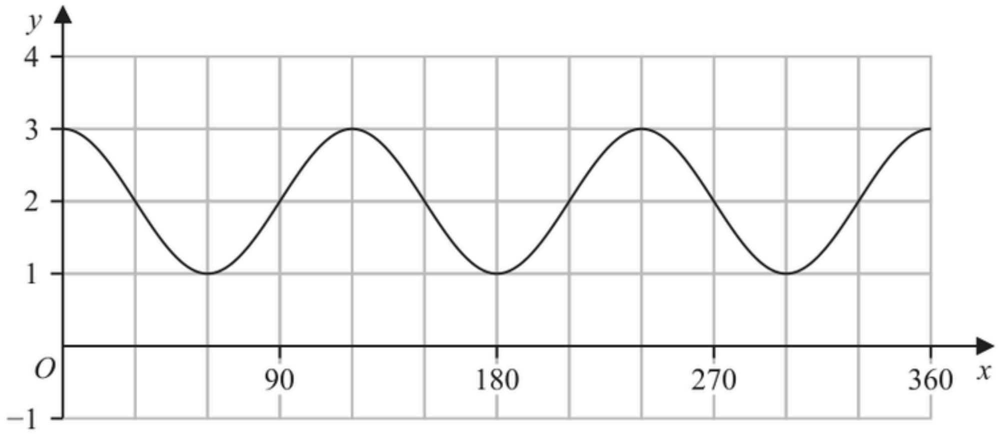
There is one minimum point on this curve.

The coordinates of this minimum point are $(8, 1)$

(a) Write down the coordinates of the minimum point on the curve with equation $y = f(4x)$

(.............................. , ...)
(1)

The diagram shows the graph of $y = \cos(ax) + b$ for $0 \leqslant x \leqslant 360$



(b) (i) Find the value of a

a = ...
(1)

(ii) Find the value of b

b = ...
(1)

(Total for Question 25 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Transformations of Graphs

18 questions

Transformations of Graphs

Q#: 27 | Marks: 4

27 A curve has equation $y = \mathrm{f}(x)$

There is one maximum point on this curve.
The coordinates of this maximum point are (6, −5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = \mathrm{f}(x - 4)$

(.......................... , ...)
(1)

(ii) $y = \mathrm{f}(3x)$

(.......................... , ...)
(1)

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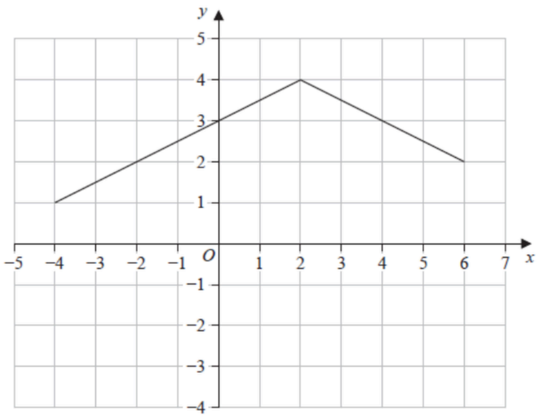
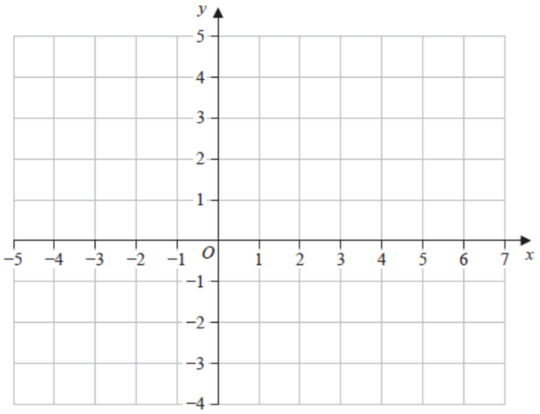
Here is the graph of $y = \mathrm{g}(x)$



(b) On the grid below, draw the graph of $y = \mathrm{g}(2x) - 3$



(2)

(Total for Question 27 is 4 marks)

CLASSIFIED TOPIC

MODULAR

Differentiation

14 questions

Differentiation

Q#: 22 Marks: 4

22 Work out the range of values of x for which the graph of $y = x^3 + 2\,x^2 - 10x + 5$
has a positive gradient.
Show clear algebraic working.

....

(Total for Question 22 is 4 marks)

Differentiation

Q#: 22 Marks: 6

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11
The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b
Show clear algebraic working.

(Total for Question 22 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Transformations of Graphs

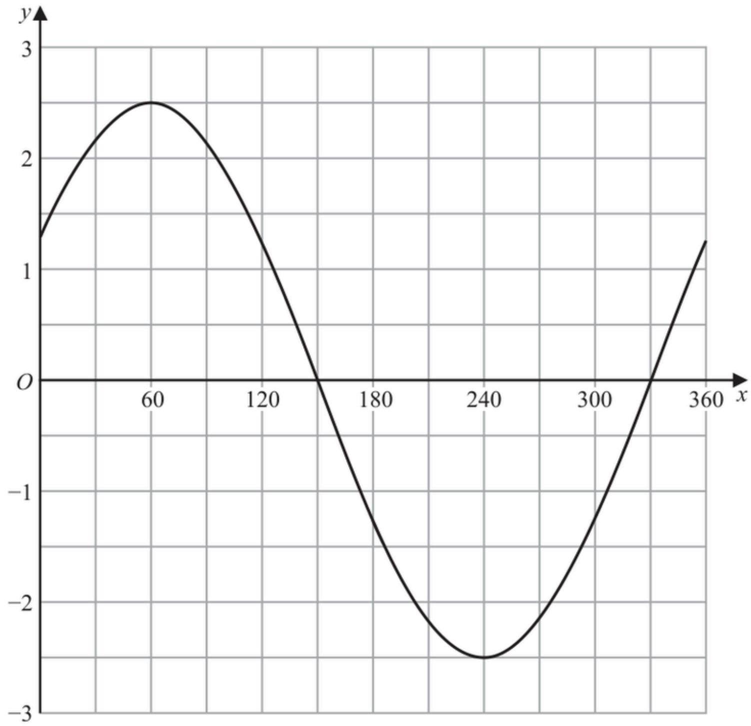
18 questions

Transformations of Graphs

Q#: 22

Marks: 4

22 The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid.



(a) Find the value of a and the value of b.

$a =$ ....

$b =$ ....

(2)

Another curve C has equation $y = \mathrm{f}(x)$

The coordinates of the minimum point of C are (4, 5)

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = \mathrm{f}(2x)$

(.................... , ...)

(ii) $y = \mathrm{f}(x) - 7$

(.................... , ...)

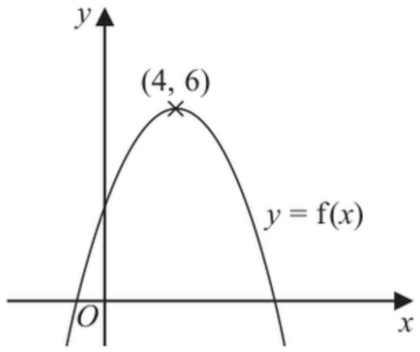
(2)

(Total for Question 22 is 4 marks)

Transformations of Graphs

Q#: 21 Marks: 4

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.............................. , ...)

(ii) $y = f(2x)$

(.............................. , ...)

(2)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

....

(2)

(Total for Question 21 is 4 marks)

Transformations of Graphs

Q#: 21 | Marks: 4

21 A curve has equation $y = f(x)$

The coordinates of the minimum point on this curve are (–9, 15)

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

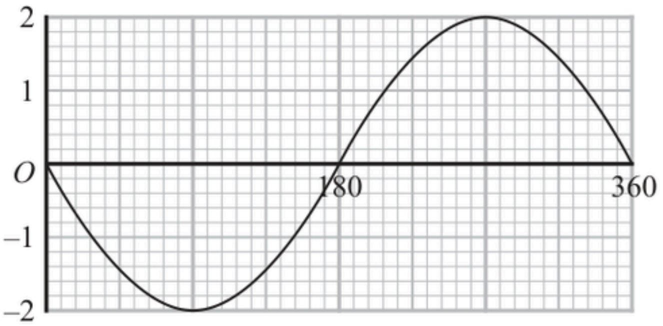
(.......................... , ...)

(ii) $y = \frac{1}{3}f(x)$

(.......................... , ...)

(2)

The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $a > 0$ and that $0 < b < 360$

(b) find the value of a and the value of b.

a = ...

b = ...

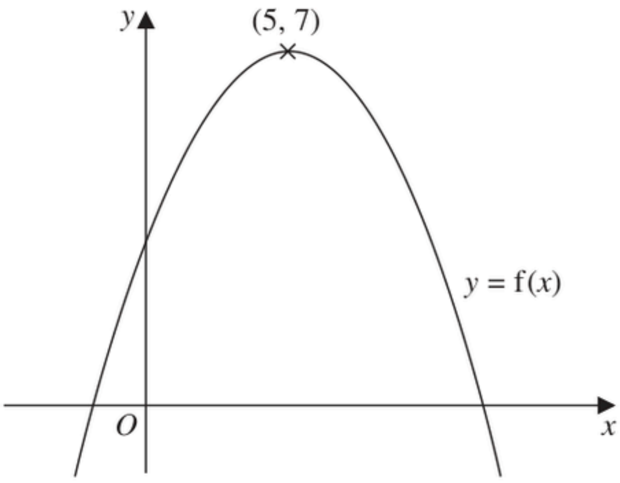
(2)

(Total for Question 21 is 4 marks)

Transformations of Graphs

Q#: 23 Marks: 2

23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(..................... , ...)

(ii) $y = f(x) + 3$

(..................... , ...)

(Total for Question 23 is 2 marks)

Transformations of Graphs

Q#: 20 Marks: 6

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x) + 5$

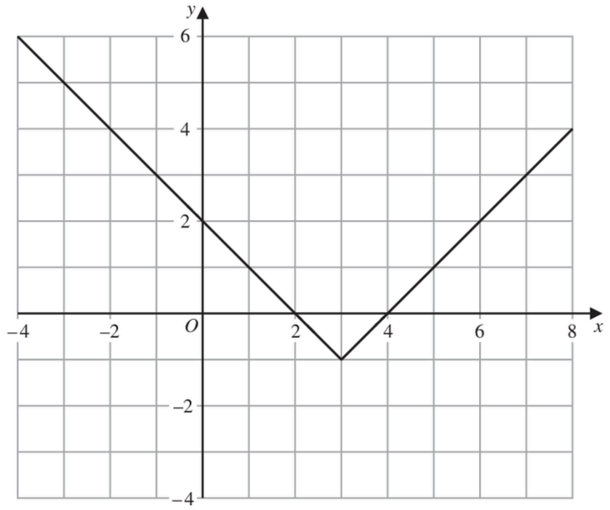
(.................... , ...)

(ii) $y = f(3x)$

(.................... , ...)

(2)

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leqslant x \leqslant 7$

(2)

The graph of $y = h(x)$ intersects the x-axis at two points.
The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

.... , ...

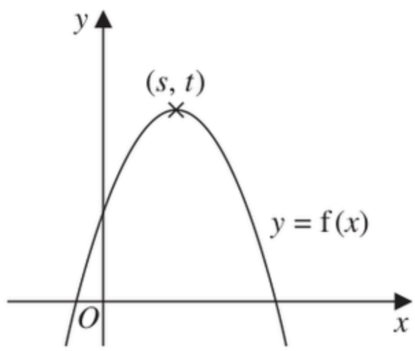
(2)

(Total for Question 20 is 6 marks)

Transformations of Graphs

Q#: 20 Marks: 2

20 The diagram shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(i) $y = \mathrm{f}(x - 2)$

(.......................... , ...)
(1)

(ii) $y = 3\mathrm{f}(x)$

(.......................... , ...)
(1)

(Total for Question 20 is 2 marks)

Transformations of Graphs

Q#: 22 Marks: 4

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = \mathrm{f}(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

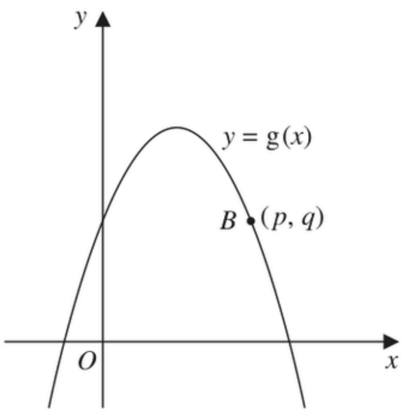
(i) $y = \mathrm{f}(x) - 3$

(.......................... , ...)
(1)

(ii) $y = \mathrm{f}\left(\dfrac{x}{2}\right)$

(.......................... , ...)
(1)

Here is a sketch of part of the curve with equation $y = \mathrm{g}(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -\mathrm{g}(x - c)$$

where c is a constant.

(.......................... , ...)
(2)

(Total for Question 22 is 4 marks)

Transformations of Graphs

Q#: 23 Marks: 2

23 A curve has equation $y = f(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.............................. , ...)
(1)

(ii) $y = f(3x)$

(.............................. , ...)
(1)

(Total for Question 23 is 2 marks)

Transformations of Graphs

Q#: 21 Marks: 3

21 The curve with equation $y = f(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

(a) Write down the coordinates of the turning point on the curve with equation $y = f(3x)$

(.............................. , ...)
(1)

The curve **C** with equation $y = g(x)$ is transformed to give the curve **S** with equation $y = g(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

(b) Write down the value of a and the value of b

$a =$ ....

$b =$ ....
(2)

(Total for Question 21 is 3 marks)

Transformations of Graphs

Q#: 21 Marks: 2

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.............................. , ...)

(1)

(b) $y = f(3x)$

(.............................. , ...)

(1)

(Total for Question 21 is 2 marks)

Transformations of Graphs

Q#: 20 | Marks: 7

20 Curve **C** has equation $y = f(x)$
The graph of curve **C** has one maximum point.
The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.......................... , ...)
(1)

(ii) $y = f(x) - 7$

(.......................... , ...)
(1)

(iii) $y = f(-x)$

(.......................... , ...)
(1)

Curve **L** has equation $y = x^2 + 7x + 20$

Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

(b) Find an equation for **S**
Give your answer in the form $y = ax^2 + bx + c$

$y =$ ....
(4)

(Total for Question 20 is 7 marks)

Transformations of Graphs

Q#: 21 Marks: 2

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.
The coordinates of this minimum point are (5, –4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(.............................. , ...)
(1)

(ii) $y = f(x) - 6$

(.............................. , ...)
(1)

(Total for Question 21 is 2 marks)

Transformations of Graphs

Q#: 25 Marks: 3

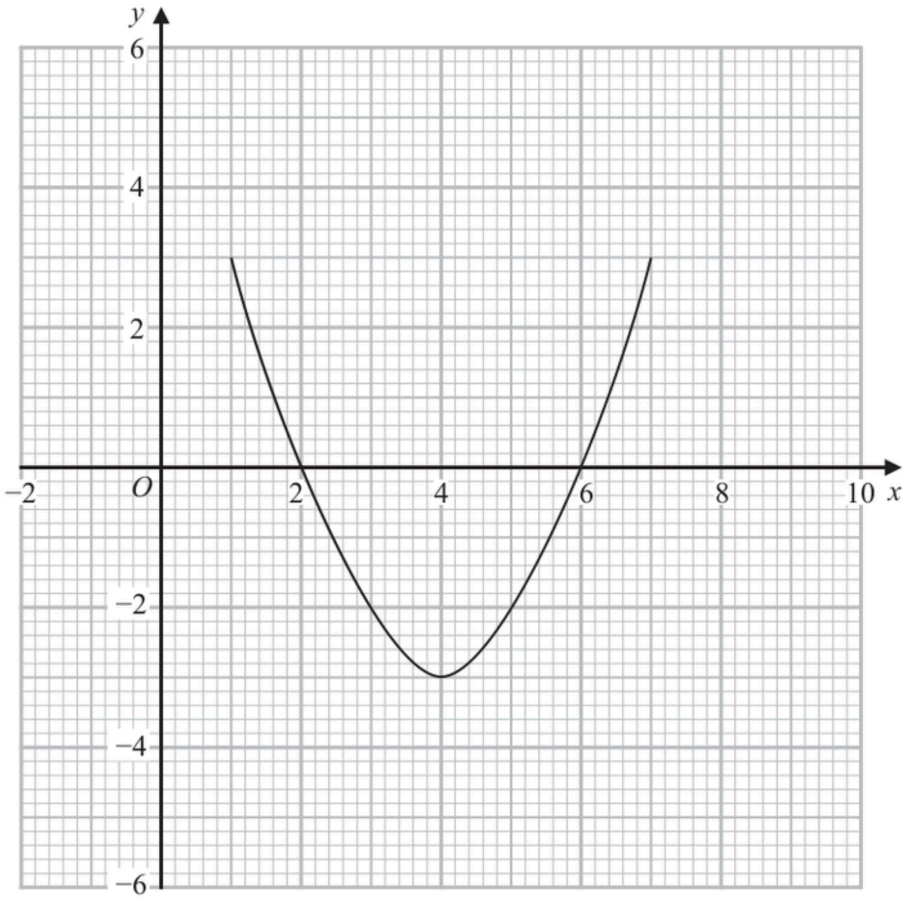
25 The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

....

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

Transformations of Graphs

Q#: 20 | Marks: 2

20 A curve has equation $y = f(x)$
There is only one maximum point on the curve.
The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(.............................. , ...)

(ii) $y = f(2x)$

(.............................. , ...)

(Total for Question 20 is 2 marks)

Transformations of Graphs

Q#: 21

Marks: 6

21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.................... , ...)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

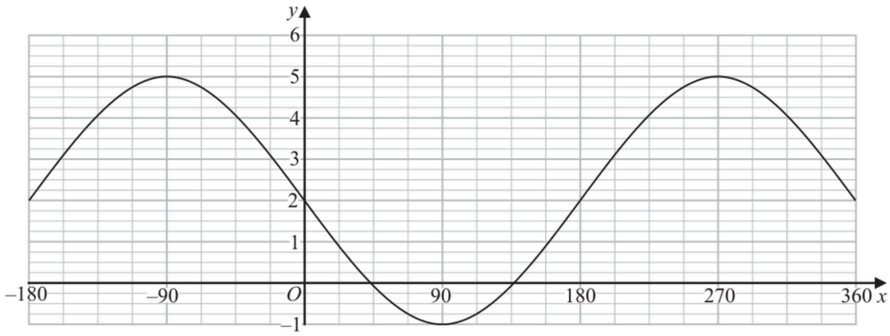
....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

....
(1)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

$a =$ ....

$b =$ ....

$c =$ ....
(3)

(Total for Question 21 is 6 marks)

Transformations of Graphs

Q#: 24 Marks: 4

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$g(x) = 5(x+4)^2 + 9(x+4) + 8$ by the translation $\begin{pmatrix} a \\ b \end{pmatrix}$

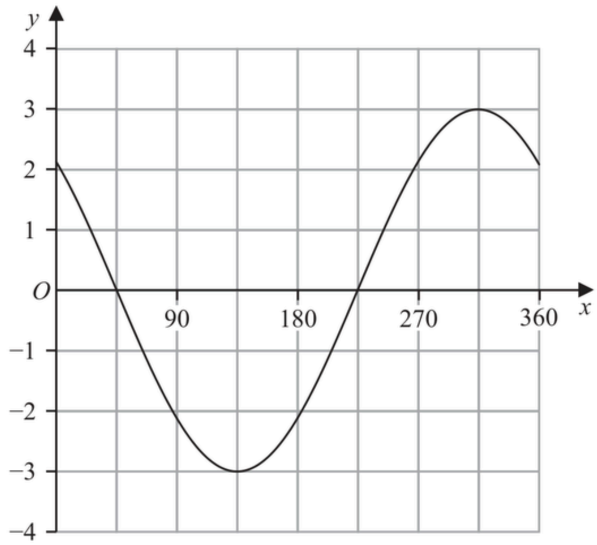
(a) Write down the value of a and the value of b

$a =$ ....

$b =$ ....

(2)

The graph of $y = p\cos(x+q)^\circ$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p =$ ....

$q =$ ....

(2)

(Total for Question 24 is 4 marks)

Transformations of Graphs

Q#: 24 Marks: 3

24 A curve has equation $y = \mathrm{f}(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = \mathrm{f}(x + 5)$

(.............................. , ...)
(1)

(ii) $y = \mathrm{f}(x) + 7$

(.............................. , ...)
(1)

(iii) $y = \mathrm{f}(2x)$

(.............................. , ...)
(1)

(Total for Question 24 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Differentiation

14 questions

Differentiation

Q#: 23

Marks: 6

23 A particle moves along a straight line.
The fixed point O lies on this line.
The displacement of the particle from O at time t seconds, $t \geqslant 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geqslant -5$

Find an expression for T in terms of V.

Give your expression in the form $\dfrac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found.

$T =$ ....

(Total for Question 23 is 6 marks)

Differentiation

Q#: 20 Marks: 5

20 A particle P is moving along a straight line.
The fixed point O lies on the line.

At time t seconds ($t \geqslant 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P.

.... m/s

(Total for Question 20 is 5 marks)

Differentiation

Q#: 24 Marks: 6

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**
The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3
Show clear algebraic working.

$y =$ ....

(Total for Question 24 is 6 marks)

Differentiation

Q#: 23 Marks: 6

23 Two particles, P and Q, move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

Differentiation

Q#: 24

Marks: 5

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geqslant 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is a m/s^2

Find the value of a

$a =$ ....

(Total for Question 24 is 5 marks)

Differentiation

Q#: 22

Marks: 7

22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

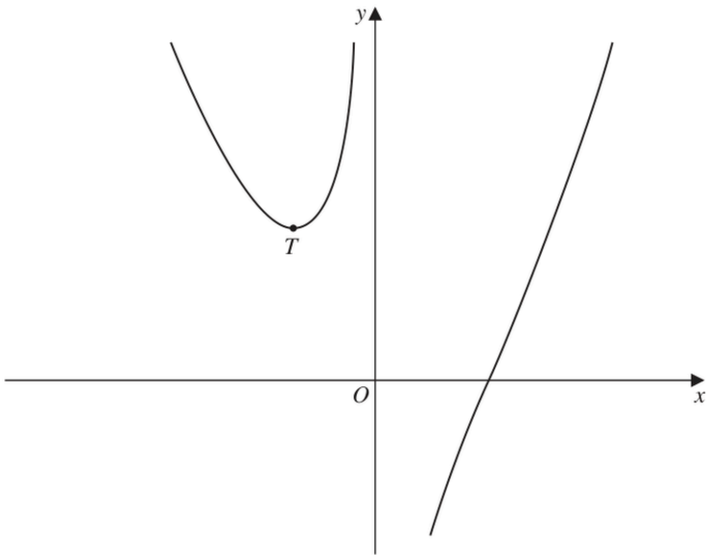


Diagram **NOT** accurately drawn

For all values of p, the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

(a) find the value of p

p = ...

(4)

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

(b) Find the value of k

k = ...

(3)

(Total for Question 22 is 7 marks)

Differentiation

Q#: 22 | Marks: 5

22 The diagram shows a solid cuboid.

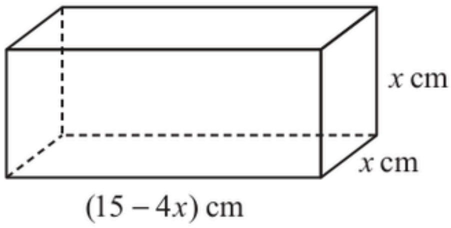


Diagram **NOT** accurately drawn

The volume of the cuboid is V cm^3

Find the maximum value of V

(Total for Question 22 is 5 marks)

Differentiation

Q#: 21 | Marks: 5

21 The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a.
Show your working clearly.

a = ...

(Total for Question 21 is 5 marks)

Differentiation

Q#: 21

Marks: 6

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

(2)

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

Differentiation

Q#: 23 | Marks: 4

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x, in terms of p and m, for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

Differentiation

Q#: 20

Marks: 4

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

(Total for Question 20 is 4 marks)

Differentiation

Q#: 22 | Marks: 6

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11
The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b
Show clear algebraic working.

(Total for Question 22 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Angles in Polygons & Parallel Lines

1 questions

Angles in Polygons & Parallel Lines

Q#: 23 Marks: 6

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n
Show clear algebraic working.

Question 23 continued

$n =$ ....

(Total for Question 23 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Bearings, Scale Drawing & Constructions

1 questions

Bearings, Scale Drawing & Constructions

Q#: 24 Marks: 6

24 A, B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A
C is on a bearing of 129° from B

Calculate the bearing of A from C
Give your answer correct to the nearest degree.

....°

(Total for Question 24 is 6 marks)

CLASSIFIED TOPIC

MODULAR

Circle Theorems

2 questions

Circle Theorems

Q#: 20 | Marks: 2

20 A, B and C are points on a circle.

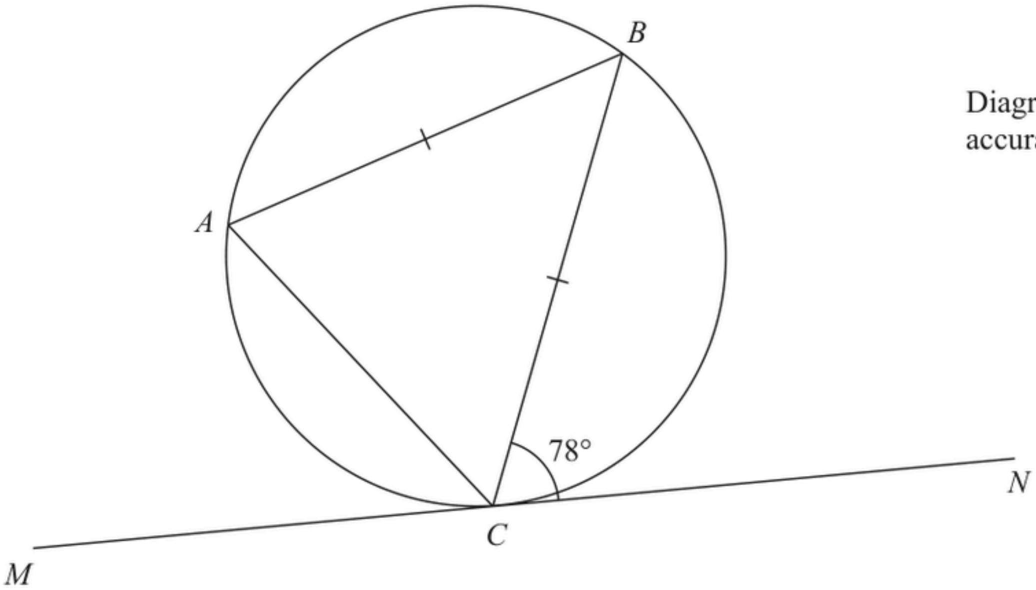


Diagram **NOT** accurately drawn

MN is the tangent to the circle at C

$AB = CB$

Angle $BCN = 78°$

Find the size of angle ABC

.... °

(Total for Question 20 is 2 marks)

Circle Theorems

Q#: 23 Marks: 5

23

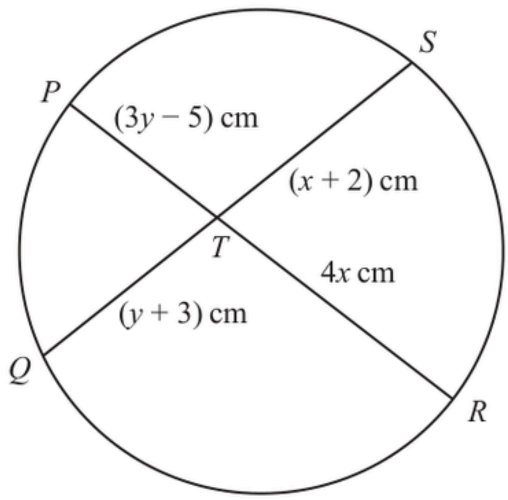


Diagram **NOT** accurately drawn

PTR and *QTS* are chords of a circle.

$PT = (3y - 5)$ cm $QT = (y + 3)$ cm $RT = 4x$ cm $ST = (x + 2)$ cm

Find an expression for y in terms of x

$y =$ ....

Total for Question 23 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Volume & Surface Area

13 questions

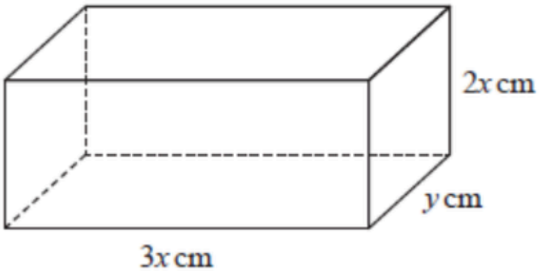
Volume & Surface Area

Q#: 23

Marks: 3

23 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is A cm^2

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

Volume & Surface Area

Q#: 21 | Marks: 5

21 The diagram shows a solid cuboid.

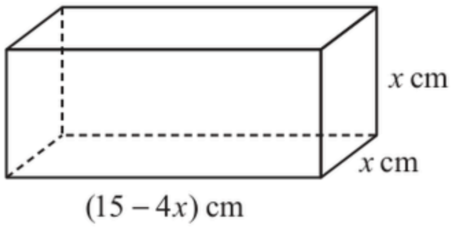


Diagram **NOT** accurately drawn

The volume of the cuboid is V cm^3

Find the maximum value of V

(Total for Question 21 is 5 marks)

Volume & Surface Area

Q#: 25 | Marks: 5

25 The diagram shows a solid hemisphere, H

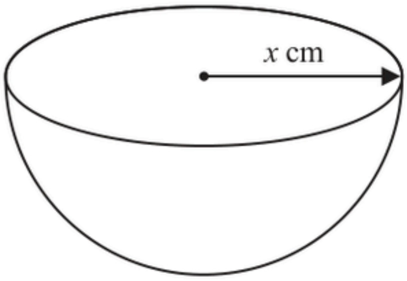


Diagram **NOT** accurately drawn

The radius of H is x cm
The volume of H is 6174π cm^3

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

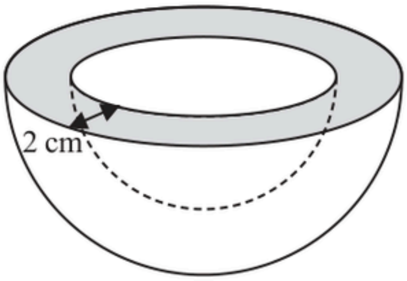


Diagram **NOT** accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

.... cm^2

(Total for Question 25 is 5 marks)

Volume & Surface Area

Q#: 22 | Marks: 5

22 A solid is made from a cone and a hemisphere.

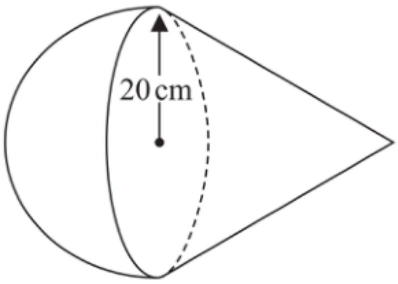


Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi\,\text{cm}^2$

The volume of the solid is $k\pi\,\text{cm}^3$

Work out the exact value of k

$k =$ ....

(Total for Question 22 is 5 marks)

CLASSIFIED TOPIC

MODULAR

Area & Volume of Similar Shapes

16 questions

Area & Volume of Similar Shapes

Q#: 25 | Marks: 4

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** − volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 25 is 4 marks)

Area & Volume of Similar Shapes

Q#: 23 Marks: 3

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm^2)	**Volume (cm^3)**
Shape **P**	200	672
Shape **Q**	450	

Work out the volume of shape **Q**

.... cm^3

(Total for Question 23 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Volume & Surface Area

13 questions

Volume & Surface Area

Q#: 26 | Marks: 6

26 Here is a sector, AOB, of a circle with centre O and angle $AOB = x°$

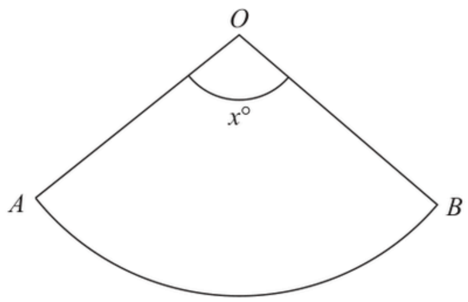


Diagram **NOT** accurately drawn

The sector can form the curved surface of a cone by joining OA to OB.

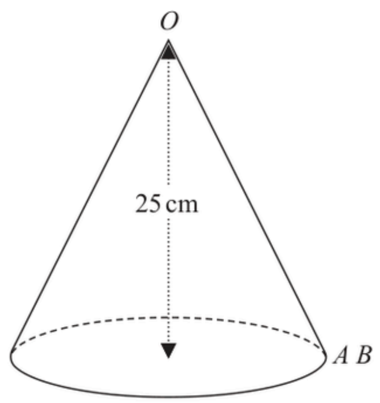


Diagram **NOT** accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x.
Give your answer correct to the nearest whole number.

$x =$ ....

(Total for Question 26 is 6 marks)

Volume & Surface Area

Q#: 22 | Marks: 5

22 A solid is made from a cone and a hemisphere.

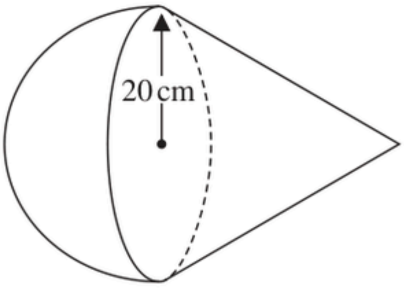


Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

k = ...

(Total for Question 22 is 5 marks)

Volume & Surface Area

Q#: 24

Marks: 5

24 The surface area of sphere **A** is nine times the surface area of sphere **B**
The difference between the volume of sphere **A** and the volume of sphere **B** is 117π cm^3

Find the radius of the smaller sphere.
Show your working clearly.

.... cm

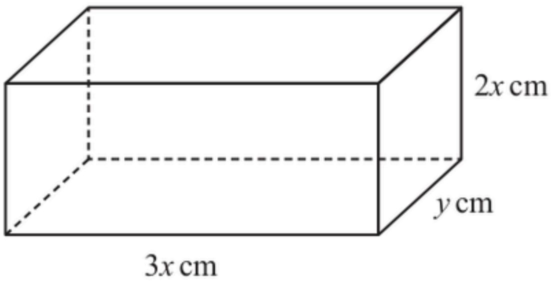
(Total for Question 24 is 5 marks)

Volume & Surface Area

Q#: 21 | Marks: 3

21 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is $1014\,\text{cm}^3$
The total surface area of the cuboid is $A\,\text{cm}^2$

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

Volume & Surface Area

Q#: 26 Marks: 5

26 The diagram shows a solid hemisphere, H

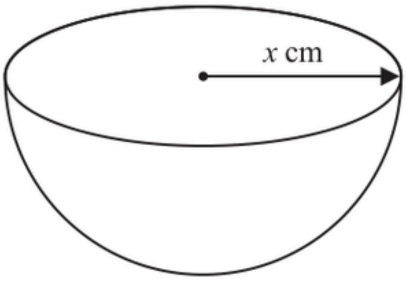


Diagram **NOT** accurately drawn

The radius of H is x cm
The volume of H is 6174π cm^3

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

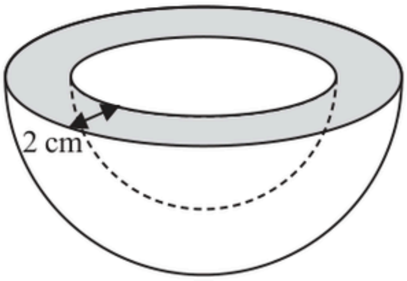


Diagram **NOT** accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

.... cm^2

(Total for Question 26 is 5 marks)

Volume & Surface Area

Q#: 23

Marks: 5

23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

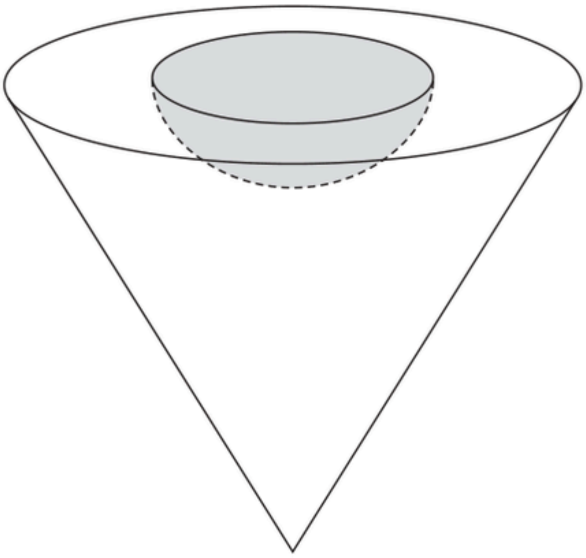


Diagram **NOT** accurately drawn

The radius of the hemisphere is $2x$ cm
The radius of the base of the cone is $5x$ cm
The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm^3

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.
Give your answer correct to the nearest integer.

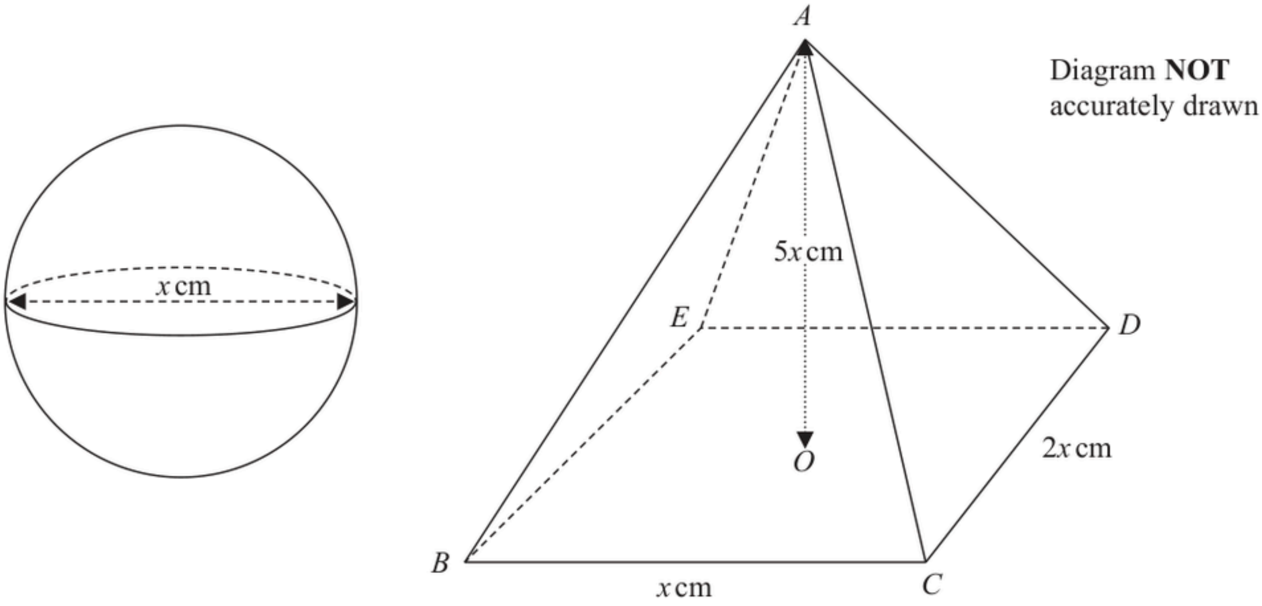
.... cm^2

(Total for Question 23 is 5 marks)

Volume & Surface Area

Q#: 22 Marks: 6

22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is $288\pi \text{ cm}^3$

Calculate the total surface area of the pyramid.
Give your answer correct to the nearest cm²

.... cm²

(Total for Question 22 is 6 marks)

Volume & Surface Area

Q#: 21 Marks: 5

21 Here is a triangular prism *ABCDEF*

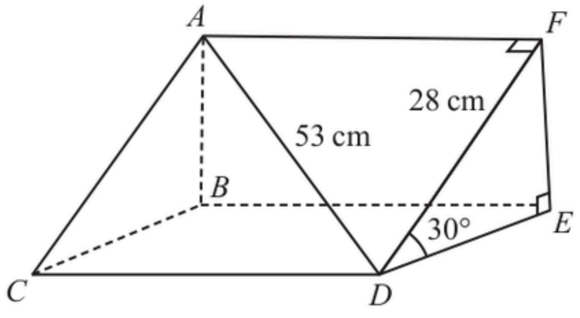


Diagram **NOT** accurately drawn

$AD = 53$ cm
$DF = 28$ cm
Angle $FDE = 30°$

Work out the volume of the triangular prism.
Give your answer correct to the nearest whole number.

.... cm^3

(Total for Question 21 is 5 marks)

Volume & Surface Area

Q#: 22

Marks: 3

22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

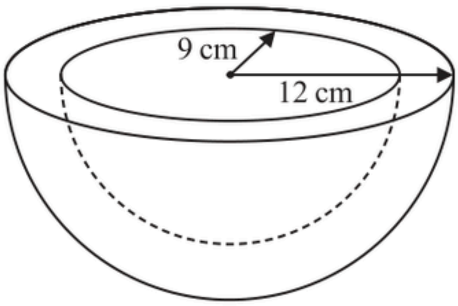


Diagram **NOT** accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

.... cm^3

(Total for Question 22 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Area & Volume of Similar Shapes

16 questions

Area & Volume of Similar Shapes

Q#: 20 | Marks: 5

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

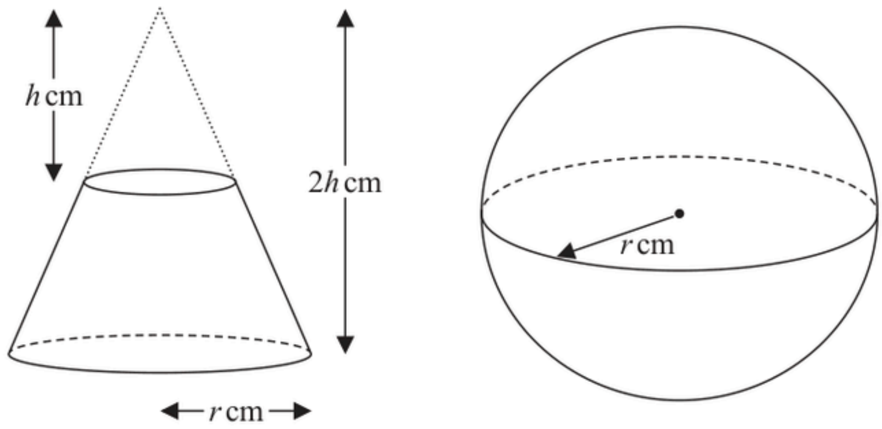


Diagram **NOT** accurately drawn

The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h.
Give your expression in its simplest form.

$r =$ ....

(Total for Question 20 is 5 marks)

Area & Volume of Similar Shapes

Q#: 20 Marks: 4

20 The diagram shows two similar vases, **A** and **B**.

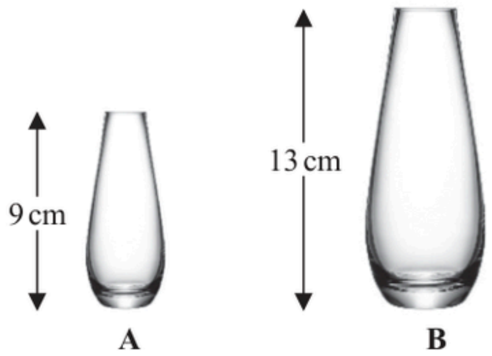


Diagram **NOT** accurately drawn

The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase } \mathbf{A} + \text{surface area of vase } \mathbf{B} = 1800\,\text{cm}^2$$

calculate the surface area of vase **A**.

.... cm^2

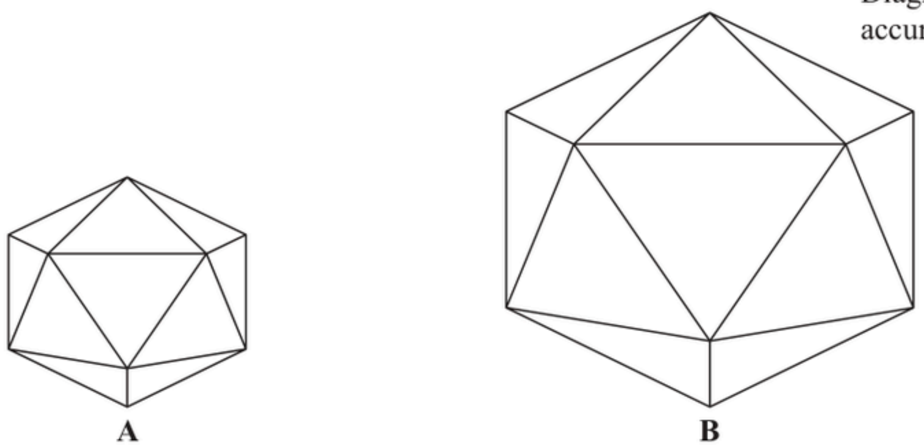
(Total for Question 20 is 4 marks)

Area & Volume of Similar Shapes

Q#: 20 | Marks: 3

20 **A** and **B** are two similar solids.

Diagram **NOT** accurately drawn



A has a volume of 1836 cm^3
B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

.... cm^2

(Total for Question 20 is 3 marks)

Area & Volume of Similar Shapes

Q#: 20 Marks: 4

20 The diagram shows two similar metal statues.

Diagram **NOT** accurately drawn

A **B**

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is k% less than the surface area of statue **A**

Work out the value of k
Give your answer correct to 3 significant figures.

k = ...

(Total for Question 20 is 4 marks)

Area & Volume of Similar Shapes

Q#: 24 Marks: 6

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.
The small cone and the large cone are similar.

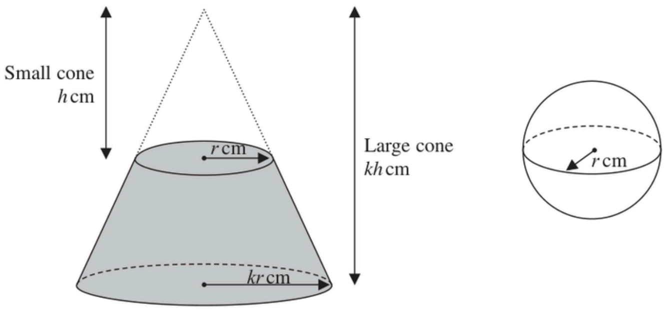


Diagram **NOT** accurately drawn

The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

The sphere is divided into two hemispheres, each of radius r cm.

Solid **A** is formed by joining one of the hemispheres to the frustum.
The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid **B** is formed by joining the other hemisphere to the small cone that was removed from the large cone.
The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.

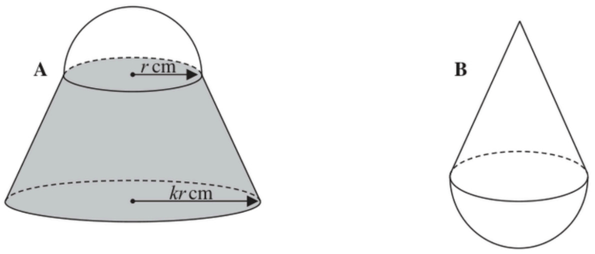


Diagram **NOT** accurately drawn

The volume of solid **A** is 6 times the volume of solid **B**.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = $ ....

(Total for Question 24 is 6 marks)

Area & Volume of Similar Shapes

Q#: 24 Marks: 5

24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

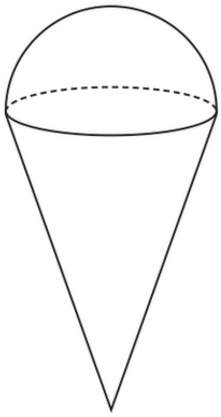


Diagram **NOT** accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is 4 × the radius of the hemisphere.

A separate sphere has radius kx cm.
The volume of this sphere is 12.5 × the volume of **S**

(a) Work out the value of k

$k =$ ....

(4)

A solid, **T**, is similar to solid **S**

The volume of **T** is 512 × the volume of **S**

The total surface area of **T** is d × the total surface area of **S**

(b) Find the value of d

$d =$ ....

(1)

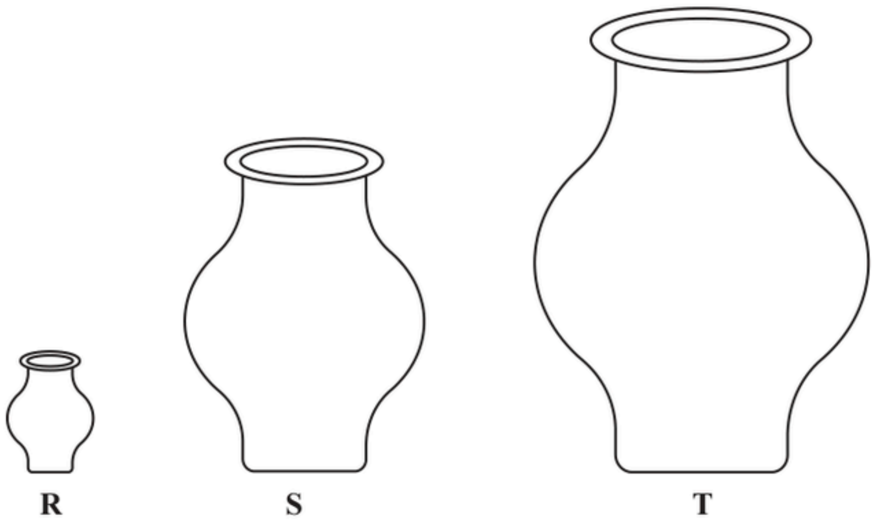
(Total for Question 24 is 5 marks)

Area & Volume of Similar Shapes

Q#: 26 | Marks: 4

26 **R**, **S** and **T** are three similar vases.

Diagram **NOT** accurately drawn



The volume of vase **S** is 72.8% more than the volume of vase **R**

The height of vase **R** is h cm
The height of vase **T** is $6h$ cm

The surface area of vase **S** is A cm^2
The surface area of vase **T** is kA cm^2

Work out the value of k

$k =$ ....

(Total for Question 26 is 4 marks)

Area & Volume of Similar Shapes

Q#: 26 Marks: 4

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** = 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

Area & Volume of Similar Shapes

Q#: 20 Marks: 3

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is $300\,\text{cm}^3$

Given that

the surface area of each small block : the surface area of each large block = 25 : 36

work out the volume of each large block.

.... cm^3

(Total for Question 20 is 3 marks)

Area & Volume of Similar Shapes

Q#: 23 Marks: 5

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.

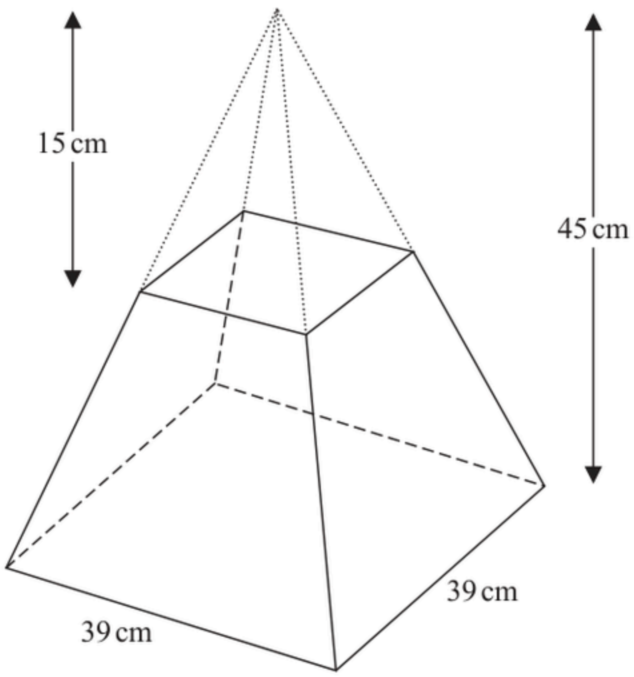


Diagram **NOT** accurately drawn

The height of the small pyramid is 15 cm.
The height of the large pyramid is 45 cm.
The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.
Give your answer correct to the nearest whole number.

.... cm^2

(Total for Question 23 is 5 marks)

Area & Volume of Similar Shapes

Q#: 20 Marks: 3

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3
Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

.... cm^3

(Total for Question 20 is 3 marks)

Area & Volume of Similar Shapes

Q#: 23 Marks: 5

23 Solid **A** is similar to solid **B**

Here is some information about solid **A** and solid **B**

	solid **A**	solid **B**
Height (cm)	3^x	
Area (cm^2)	7776	486
Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid **B**
Give your answer as a decimal.

.... cm

(Total for Question 23 is 5 marks)

Area & Volume of Similar Shapes

Q#: 23 Marks: 5

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

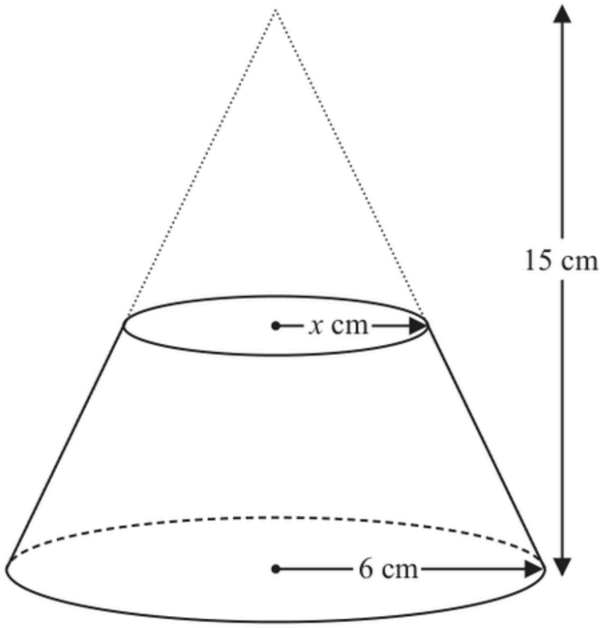


Diagram **NOT** accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.
The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

x = ...

(Total for Question 23 is 5 marks)

Area & Volume of Similar Shapes

Q#: 23 Marks: 3

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm^2)	**Volume (cm^3)**
Shape **P**	200	672
Shape **Q**	450	

Work out the volume of shape **Q**

.... cm^3

(Total for Question 23 is 3 marks)

CLASSIFIED TOPIC

MODULAR

Vectors

17 questions

Vectors

Q#: 26 Marks: 5

26

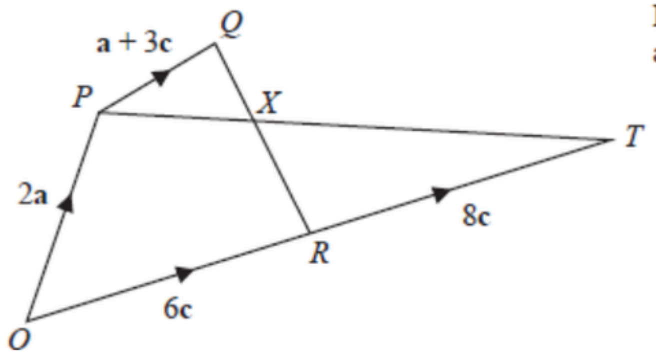


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral *OPQR* in which

$\overrightarrow{OP} = 2\mathbf{a}$ $\overrightarrow{OR} = 6\mathbf{c}$ $\overrightarrow{PQ} = \mathbf{a} + 3\mathbf{c}$

(a) Find in terms of **a** and **c**

$\overrightarrow{QR}$ = ...

(1)

The point T is such that $\overrightarrow{RT} = 8\mathbf{c}$
The point X lies on QR such that PXT is a straight line.

(b) Using a vector method, find the ratio $QX : XR$
Give your answer in the form $m : n$ where m and n are integers.
Show your working clearly.

P81641A

$QX : XR$ = ...

(4)

(Total for Question 26 is 5 marks)

Vectors

Q#: 24

Marks: 6

24 The diagram shows a quadrilateral *OABC*

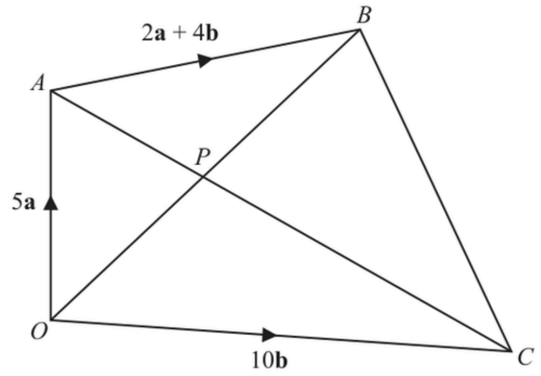


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 5\mathbf{a}$ $\overrightarrow{OC} = 10\mathbf{b}$ $\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**

$\overrightarrow{AC} =$ ....

(1)

(b) Find $\overrightarrow{OB}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{OB} =$ ....

(1)

The point *P* is such that *APC* and *OPB* are straight lines.

(c) Using a vector method, find $\overrightarrow{OP}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OP} =$ ....

(4)

(Total for Question 24 is 6 marks)

Vectors

Q#: 23

Marks: 5

23

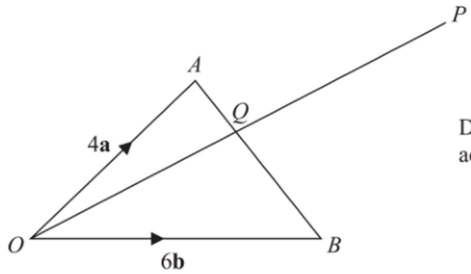


Diagram **NOT** accurately drawn

OAB is a triangle.
Q is the point on AB such that OQP is a straight line.

$$\overrightarrow{OA} = 4\mathbf{a} \qquad \overrightarrow{OB} = 6\mathbf{b} \qquad \overrightarrow{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$AQ : QB =$ ....

(Total for Question 23 is 5 marks)

Vectors

Q#: 23 Marks: 3

23

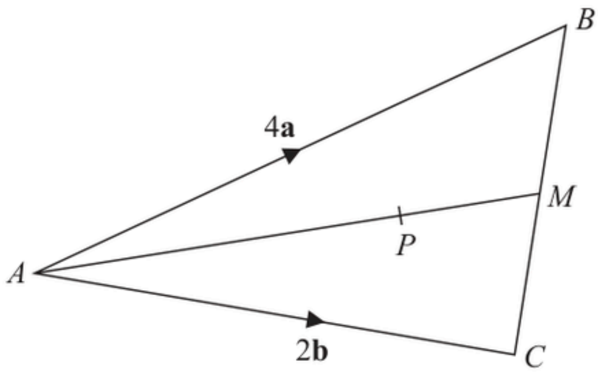


Diagram **NOT** accurately drawn

ABC is a triangle.
The midpoint of BC is M.
P is a point on AM.

$\overrightarrow{AB} = 4\mathbf{a}$

$\overrightarrow{AC} = 2\mathbf{b}$

$\overrightarrow{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$

Find the ratio $AP : PM$

(Total for Question 23 is 3 marks)

Vectors

Q#: 26 Marks: 5

26 The diagram shows trapezium $OACB$.

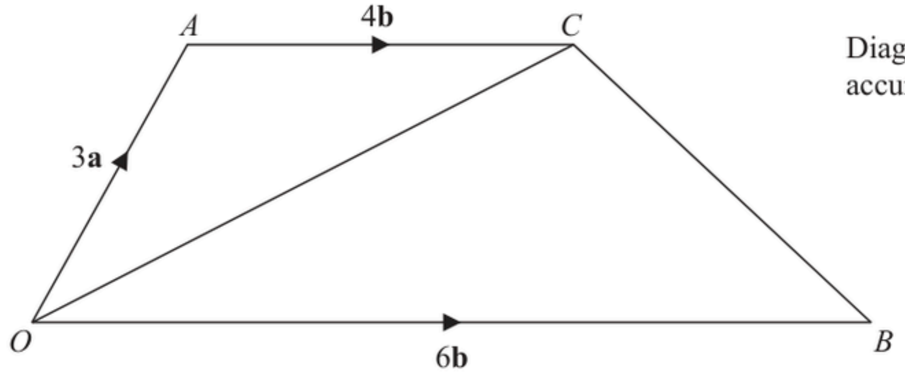


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{AC} = 4\mathbf{b}$

N is the point on OC such that ANB is a straight line.

Find $\overrightarrow{ON}$ as a simplified expression in terms of **a** and **b**.

(Total for Question 26 is 5 marks)

Vectors

Q#: 23 Marks: 6

23 OAB is a triangle.

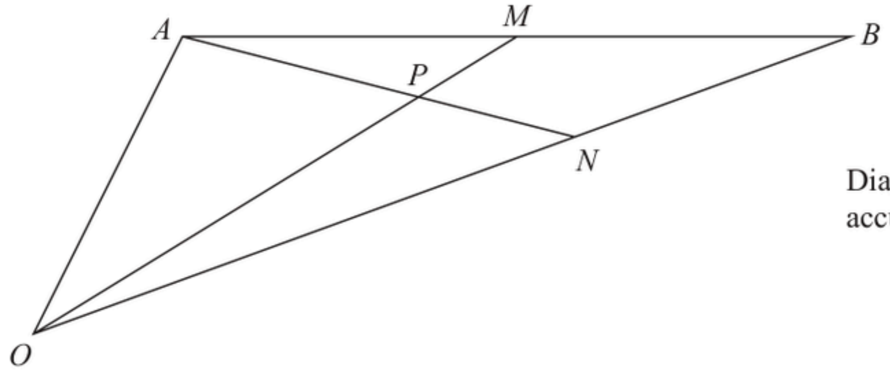


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$

M is the midpoint of AB.
N is the point on OB such that $ON : NB = 2 : 1$

P is the point on AN such that OPM is a straight line.

Use a vector method to find $OP : PM$
Show your working clearly.

(Total for Question 23 is 6 marks)

Vectors

Q#: 22 Marks: 5

22 The diagram shows triangle OAB

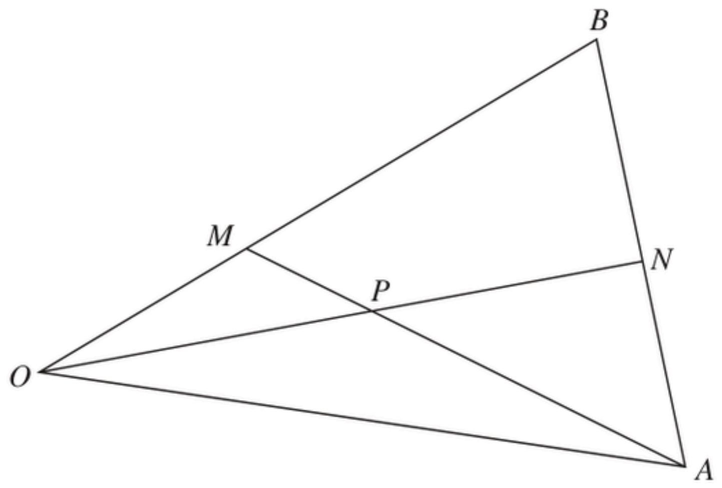


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 8\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$

M is the point on OB such that $OM : MB = 1 : 2$
N is the midpoint of AB
P is the point of intersection of ON and AM

Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**
Show your working clearly.

$\overrightarrow{OP} =$ ....

(Total for Question 22 is 5 marks)

Vectors

Q#: 24 | Marks: 5

24 $OXYZ$ is a trapezium.

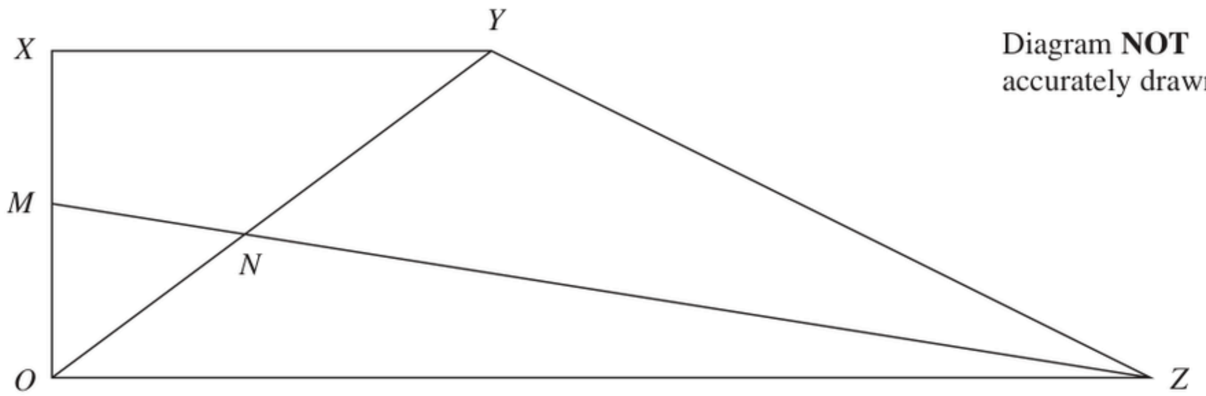


Diagram **NOT** accurately drawn

$\overrightarrow{OX} = \mathbf{a}$

$\overrightarrow{XY} = \mathbf{b}$

$\overrightarrow{OZ} = 3\mathbf{b}$

M is the midpoint of OX
N is the point such that MNZ and ONY are straight lines.

Given that $ON : OY = \lambda : 1$

use a vector method to find the value of λ

$\lambda =$ ....

(Total for Question 24 is 5 marks)

Vectors

Q#: 21 Marks: 5

21

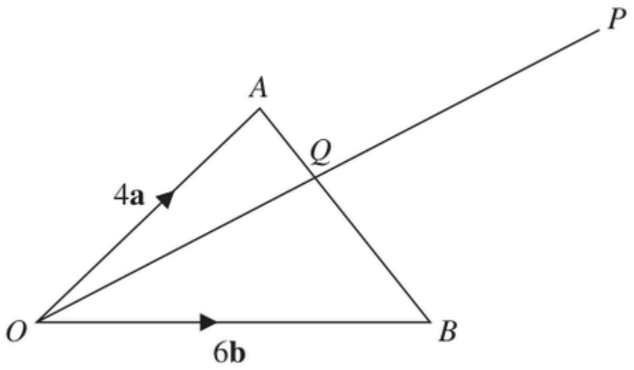


Diagram **NOT** accurately drawn

OAB is a triangle.
Q is the point on *AB* such that *OQP* is a straight line.

$\overrightarrow{OA} = 4\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{AP} = 2\mathbf{a} + 8\mathbf{b}$

Using a vector method, find the ratio $AQ : QB$

$AQ : QB =$ ....

(Total for Question 21 is 5 marks)

Vectors

Q#: 25 Marks: 5

25 *ABCD* is a parallelogram and *ADM* is a straight line.

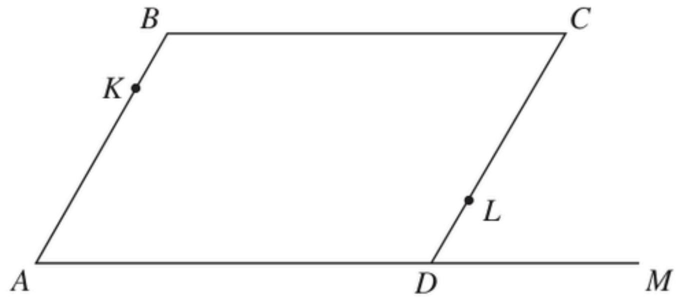


Diagram **NOT** accurately drawn

$\overrightarrow{AB} = \mathbf{a}$ $\quad \overrightarrow{BC} = \mathbf{b}$ $\quad \overrightarrow{DM} = \frac{1}{2}\mathbf{b}$

K is the point on AB such that $AK:AB = \lambda:1$
L is the point on CD such that $CL:CD = \mu:1$
KLM is a straight line.

Given that $\lambda:\mu = 1:2$

use a vector method to find the value of λ and the value of μ

$\lambda =$ ....

$\mu =$ ....

(Total for Question 25 is 5 marks)

Vectors

Q#: 24 Marks: 6

24

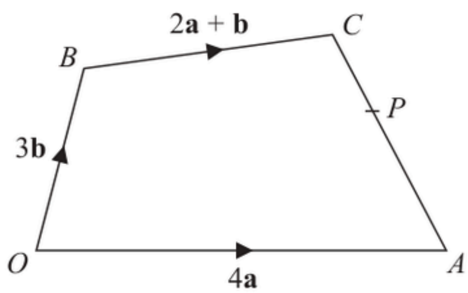


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad \overrightarrow{OB} = 3\mathbf{b}$ $\quad \overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{AC}$ = ...

(2)

The point P lies on AC such that $AP : PC = 3 : 2$

The point Q is such that OPQ and BCQ are straight lines.

(b) Using a vector method, find $\overrightarrow{OQ}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OQ}$ = ...

(4)

(Total for Question 24 is 6 marks)

Vectors

Q#: 25 | Marks: 6

25 *OPQR* is a parallelogram.

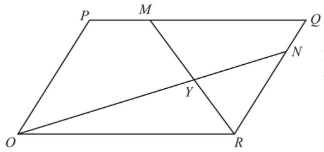


Diagram **NOT** accurately drawn

$\overrightarrow{OP} = 2\mathbf{a}$ and $\overrightarrow{OR} = 3\mathbf{b}$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of **a** and **b**, giving your answers in simplest form

(i) $\overrightarrow{ON}$

.... **(1)**

(ii) $\overrightarrow{MR}$

.... **(1)**

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ .... **(4)**

(Total for Question 25 is 6 marks)

Vectors

Q#: 23 | Marks: 5

23

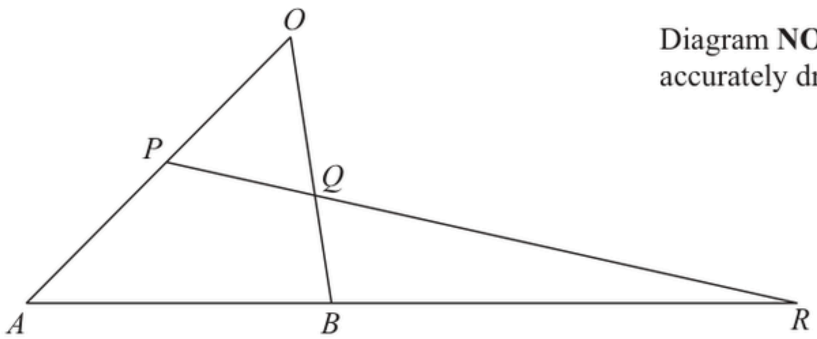


Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of OA
Q is a point on OB

ABR and PQR are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\qquad \overrightarrow{OB} = 8\mathbf{b}$

(a) Express $\overrightarrow{AB}$ in terms of **a** and **b**

....

(1)

$AB : BR = 1 : 2$ $\qquad \overrightarrow{OQ} = n\mathbf{b}$

(b) Use a vector method to find the value of n

$n =$ ....

(4)

(Total for Question 23 is 5 marks)

Vectors

Q#: 26 Marks: 5

26 *OACB* is a trapezium.

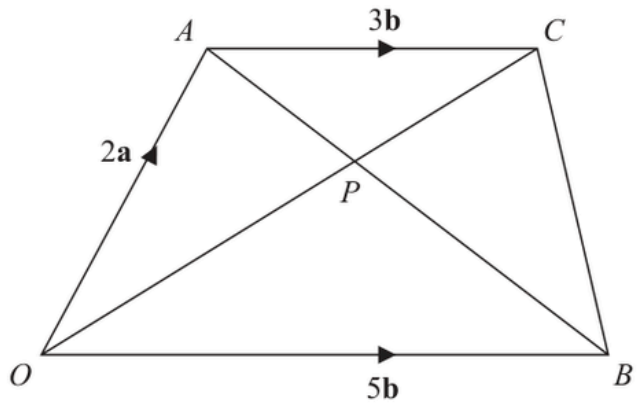


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ $\quad \overrightarrow{OB} = 5\mathbf{b}$ $\quad \overrightarrow{AC} = 3\mathbf{b}$

The diagonals, *OC* and *AB*, of the trapezium intersect at the point *P*.

Find and simplify an expression, in terms of **a** and **b**, for $\overrightarrow{OP}$
Show your working clearly.

$\overrightarrow{OP} =$ ....

(Total for Question 26 is 5 marks)

Vectors

Q#: 22 Marks: 5

22

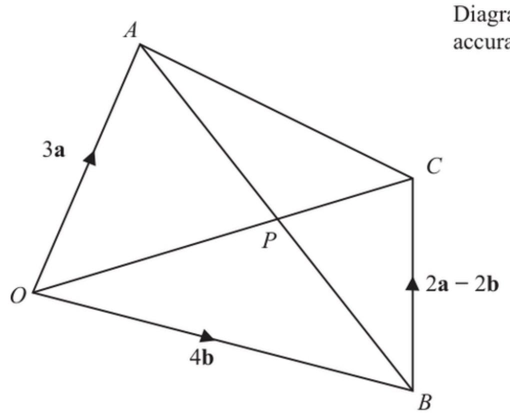


Diagram **NOT** accurately drawn

OACB is a quadrilateral.

$\overrightarrow{OA} = 3\mathbf{a}$ $\quad\overrightarrow{OB} = 4\mathbf{b}$ $\quad\overrightarrow{BC} = 2\mathbf{a} - 2\mathbf{b}$

(a) (i) Find the vector $\overrightarrow{OC}$ in terms of **a** and **b**
Simplify your answer.

$\overrightarrow{OC}$ = ...
(1)

(ii) Find the vector $\overrightarrow{AB}$ in terms of **a** and **b**

$\overrightarrow{AB}$ = ...
(1)

The point *P* lies on *AB* and on *OC*

(b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

....
(3)

(Total for Question 22 is 5 marks)

Vectors

Q#: 24 Marks: 6

24 OAB is a triangle.

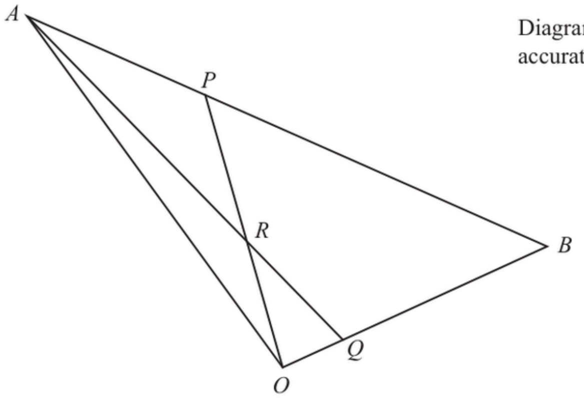


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 10\mathbf{a}$ $\overrightarrow{OB} = 10\mathbf{b}$

ARQ and ORP are straight lines.

$\overrightarrow{AP} = \frac{1}{4}\overrightarrow{AB}$ and $\overrightarrow{OQ} = \frac{1}{5}\overrightarrow{OB}$

Write the following vectors in terms of **a** and **b**
Simplify your answers.

(i) $\overrightarrow{AQ}$

....

(1)

(ii) $\overrightarrow{OP}$

....

(1)

(iii) $\overrightarrow{OR}$

....

(4)

(Total for Question 24 is 6 marks)

Vectors

Q#: 27 Marks: 6

27

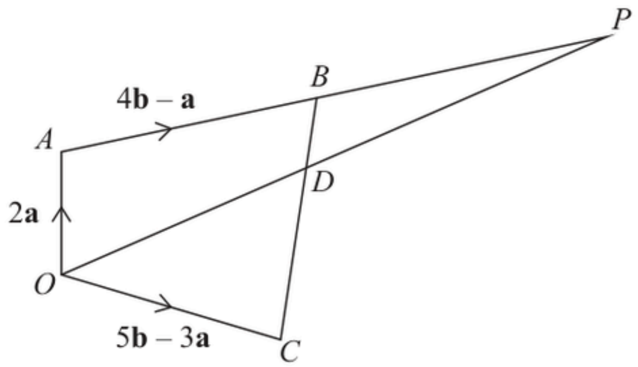


Diagram **NOT** accurately drawn

OABC is a quadrilateral.
ABP and *ODP* are straight lines.

$\overrightarrow{OA} = 2\mathbf{a}$ $\overrightarrow{AB} = 4\mathbf{b} - \mathbf{a}$ $\overrightarrow{OC} = 5\mathbf{b} - 3\mathbf{a}$

(a) Find an expression in terms of **a** and **b** for the vector $\overrightarrow{BC}$
Simplify your answer.

....
(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\overrightarrow{OP} = n\overrightarrow{OD}$

(b) use a vector method to find the value of n

$n =$ ....
(4)

(Total for Question 27 is 6 marks)